DERIVED TENSOR WITH A PSEUDO-COHERENT COMPLEX COMMUTES WITH DERIVED LIMITS OF MODULES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Rlim M_n sits in the defining distinguished triangle Rlim M_n -> prod M_n -> prod M_n -> Rlim M_n[1]. invoked at proof lines 29853-29857. Fixes the triangle whose terms get identified after tensoring with K.
- Tensoring a pseudo-coherent complex K with a product of modules computes as the derived tensor with each factor: K tensor prod M_n = prod (K tensor^L M_n). invoked at proof lines 29857-29858. Identifies the middle terms of the tensored triangle, forcing the cone to be K tensor Rlim M_n.

Source reference. Original lemma 0CYB: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16732–16732. Cited at proof lines 29857-29858 to identify the middle terms of the tensored defining triangle; load-bearing inside the counted span.

Lemma 0.2. *Let R be a ring. Let $K \in D(R)$ be pseudo-coherent. Let (M_n) be an inverse system of R-modules. Then $R\lim K \otimes^{\mathbf{L}}_R M_n = K \otimes^{\mathbf{L}}_R R\lim M_n$.*

Proof. Consider the defining distinguished triangle

$$R\lim M_n \to \prod M_n \to \prod M_n \to R\lim M_n[1]$$

and apply Lemma 0CYB. □

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